

Department of Chemical Engineering
Chemical Reaction Engineering II (CL24303)

MO2026

Tutorial 3

Assigned on	01.09.2026	Submission Date and Time	12.09.2026 23:59
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Although it is strongly suggested that you attempt all, you may solve any three of Qs1-5 for credit. Q6 is, however, mandatory

1. Show that the rate laws for irreversible surface reactions limited cases, are given by (Refer to Table 10-5 of HS Fogler):

(i)

Single site -

$$A.S \rightarrow B.S$$

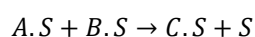
$$-r'_A = \frac{kp_A}{1 + K_A p_A + K_B p_B}$$

(ii)

Dual Site -

$$A.S + S \rightarrow B.S + S$$

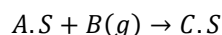
$$-r'_A = \frac{kp_A}{(1 + K_A p_A + K_B p_B)^2}$$



$$-r'_A = \frac{kp_A p_B}{(1 + K_A p_A + K_B p_B + K_C p_C)^2}$$

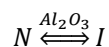
(iii)

Eley-Rideal-



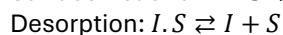
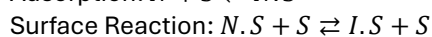
$$-r'_A = \frac{kp_A p_B}{(1 + K_A p_A + K_C p_C)}$$

2. n-pentene (*N*) isomerizes to i-pentene (*I*) over alumina (Refer to Table 10-4 of HS Fogler)



(i)

If the mechanism for the reaction follows the scheme given below:



Show that the rate law for surface-reaction-limited isomerisation is given by:

$$-r'_N = \frac{k(p_N - p_I/K_p)}{(1 + K_N p_N + K_I p_I)^2}$$

Here, $K_p = \frac{K_N K_S}{K_I}$

- (ii) If instead the mechanism for the reaction follows the scheme given below:

Adsorption: $N + S \rightleftharpoons N \cdot S$

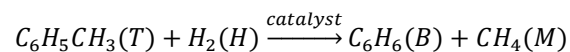
Surface Reaction: $N \cdot S \rightleftharpoons I \cdot S$

Desorption: $I \cdot S \rightleftharpoons I + S$

Show that the rate law for surface-reaction-limited isomerisation is given by:

$$-r'_N = \frac{k(p_N - p_I/K_p)}{(1 + K_N p_N + K_I p_I)}$$

- Problem **P10-3** on Page 482 of HS Fogler
- Problem **P10-5** on Page 483 of HS Fogler
- Problem **P10-6** on Page 483 of HS Fogler
- Hydrogen and toluene are reacted over crystalline silica-alumina to produce methane and benzene (Refer to section 10.4 of HS Fogler)



In a laboratory experiment in a differential reactor, the following data were obtained

Run	$-r_T \times 10^{10}$ (gmol T/gcat.s)	Partial pressure (atm)			
		p_T	p_H	p_M	p_B
1	71	1	1	1	0
2	71.3	1	1	4	0
3	41.6	1	1	0	1
4	19.7	1	1	0	4
5	42	1	1	1	1
6	17.1	1	1	0	5
7	71.8	1	1	0	0
8	142	1	2	0	0
9	284	1	4	0	0
10	47	0.5	1	0	0
11	71.3	1	1	0	0
12	117	5	1	0	0
13	127	10	1	0	0
14	131	15	1	0	0
15	133	20	1	0	0
16	41.8	1	1	1	1

- Develop a rate law consistent with experimental observations
- Propose a rate law consistent with a mechanism
- Using non-linear regression, determine model parameters – kinetic rate constant (k), equilibrium adsorption constant (K_B , K_T), and show that the rate law is given by (Example 10-1 of HS Fogler)

$$-r'_t = \frac{8.7 \times 10^{-4} p_T p_{H_2}}{1 + 1.39 p_B + 1.038 p_T} \frac{\text{mol Toluene}}{\text{g-cat.s}}$$